CLINICAL · MOUTH & THROAT

Cleft lip and palate and velopharyngeal insufficiency

Cleft lip and palate is the commonest congenital malformation of the head and neck, and its care lasts from the antenatal scan to the end of growth. The ENT surgeon's part is large and long: feeding and airway in the newborn, glue ear in nearly every child with a cleft palate, speech and velopharyngeal insufficiency, and the submucous cleft that is missed until the adenoids come out. Care is given by a multidisciplinary team, to a timetable.

AT A GLANCE

- Tensor veli palatini is the only palatal muscle supplied by V3.
- Submucous cleft: bifid uvula, zona pellucida, palpable notch.
- Lip repair at about 3 months, palate at 6 to 12 months.
- Pharyngeal flap for central gaps; sphincter pharyngoplasty for lateral gaps.

1 CHAPTER ONE Basics

Anatomy, embryology, causes.

1.1 The lip

The lip is built around the *orbicularis oris*, the sphincter of the mouth supplied by the facial nerve, with the superior labial artery running deep to it. Its surface landmarks are what the surgeon must restore: the *white roll* at the vermilion border, where a step of only 1 mm is visible; *Cupid's bow*; and the two *philtral columns*, the lines of embryological fusion, with the philtral dimple between them.

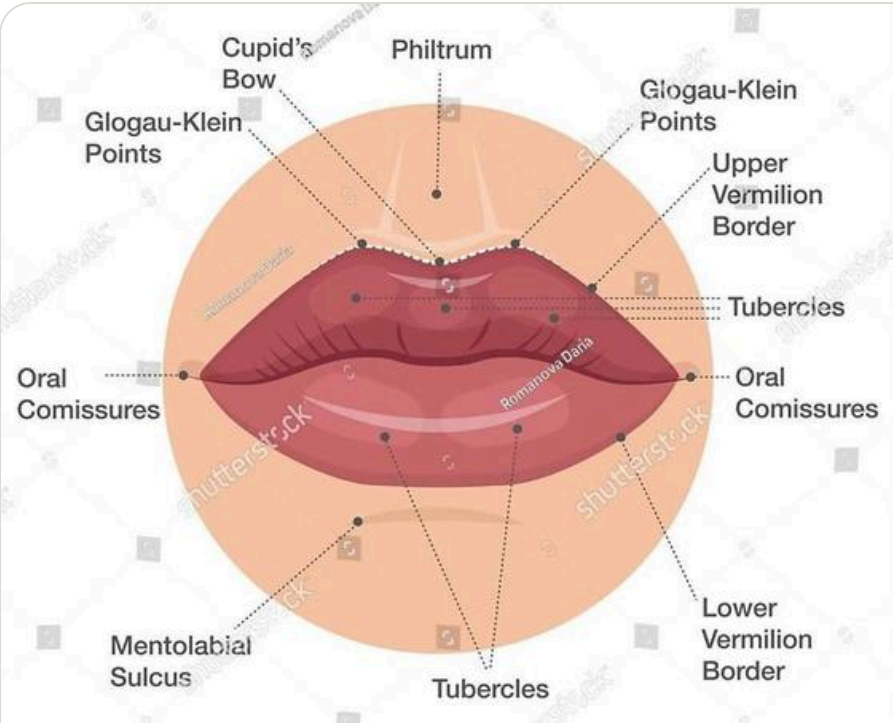


FIGURE 1 Lip landmarks: vermilion, philtrum, Cupid's bow and commissures.
Anatomy atlas · ENT Steps p. 166

1.2 The palate

The *incisive foramen* divides the palate in two. In front of it lies the **primary palate**: the premaxilla with the four incisors, the alveolus and the lip. Behind it lies the **secondary palate**: the hard palate, formed by the palatine processes of the maxillae and the palatine bones, and the soft palate.

In the soft palate, the *levator veli palatini* forms a sling that lifts the velum against the posterior pharyngeal wall, and the *tensor veli palatini* opens the Eustachian tube. The tensor is the only palatal muscle supplied by *V3*; the levator, palatoglossus, palatopharyngeus and musculus uvulae are all supplied by the vagus through the pharyngeal plexus. The hard palate is supplied by the greater palatine artery through the greater palatine foramen, the pedicle of every palatal flap, and the soft palate by the lesser palatine, ascending palatine and ascending pharyngeal arteries.

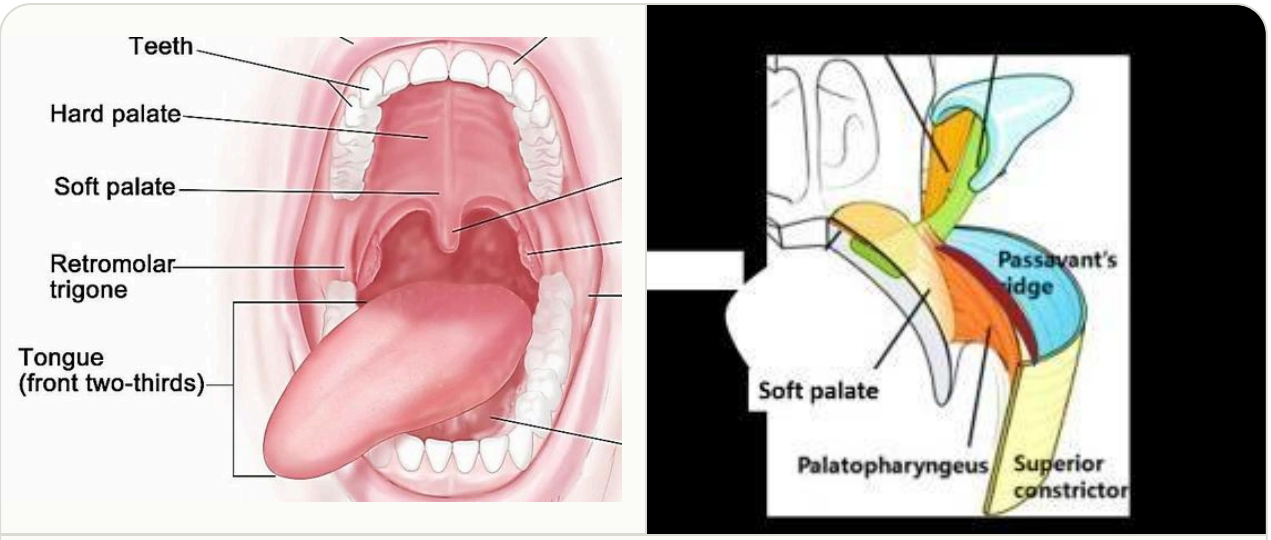


FIGURE 2 The hard and soft palate, and Passavant ridge where the velum closes against the pharyngeal wall.

Anatomy atlas · ENT Steps p. 172, ENT Steps p. 205

1.3 Embryology

The lip and primary palate form between the fourth and seventh weeks. The paired *maxillary prominences* of the first arch grow medially to fuse with the *medial nasal prominences* of the frontonasal process, and the medial nasal prominences join to form the intermaxillary segment, which becomes the philtrum, premaxilla and four incisors. Failure of fusion causes *cleft lip*, with or without a cleft alveolus.

The secondary palate forms between the sixth and twelfth weeks. The *palatine shelves* grow down beside the tongue, rise to the horizontal as the tongue drops, and fuse from front to back, from the incisive foramen to the uvula. Failure causes *cleft palate*. If the tongue cannot drop, as with a small mandible, the shelves cannot meet and the wide U-shaped cleft of the *Pierre Robin sequence* results.

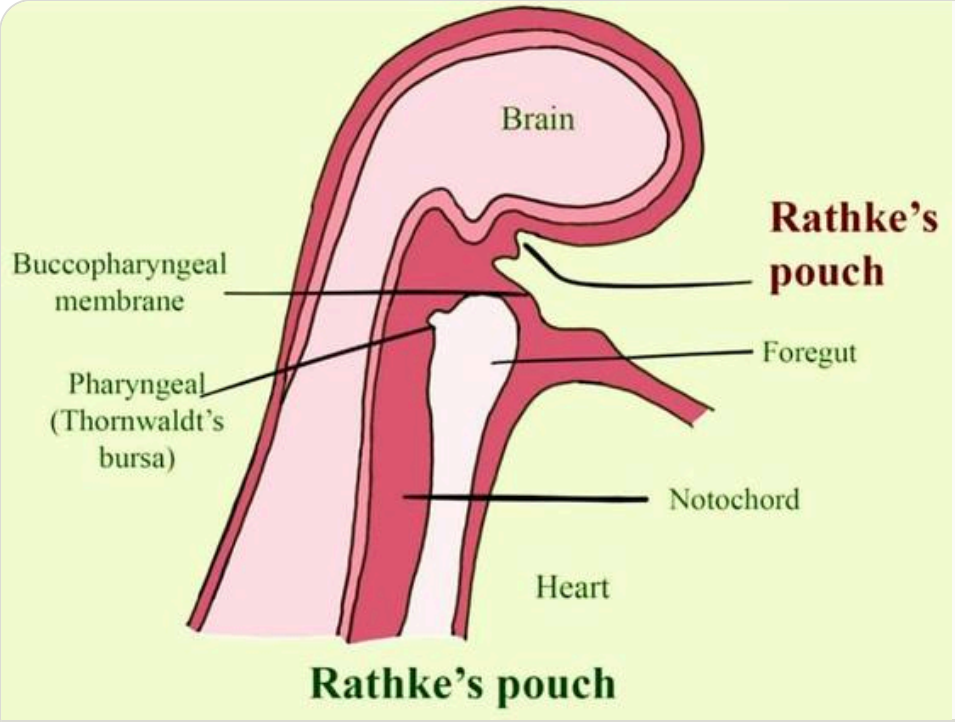


FIGURE 3 Rathke pouch and the buccopharyngeal membrane in the early embryo.
Anatomy atlas · ENT Steps p. 202

1.4 Epidemiology and causes

TWO DIFFERENT CONDITIONS

FEATURE	CLEFT LIP WITH OR WITHOUT PALATE	ISOLATED CLEFT PALATE
Frequency	About 1 in 700 to 1000 births; varies with ethnicity (commoner in Asian and Native American, less in African populations)	About 1 in 1500 to 2000; uniform across ethnicities
Sex	Boys more than girls	Girls more than boys
Associated syndromes	Up to about 30%	Up to about 50%
Side	80% unilateral, two-thirds of these on the left	—

Clefting is multifactorial. The risk to a sibling is about 4% and rises as more relatives are affected. Smoking and alcohol, anticonvulsants such as phenytoin and valproate, retinoids, steroids, folate deficiency and amniotic bands all contribute.

Several syndromes must be recognised. **Van der Woude syndrome**, autosomal dominant through IRF6, combines *lower lip pits* with cleft lip or palate and is the

commonest syndromic cleft. The **Pierre Robin sequence** runs from micrognathia to glossoptosis to a wide U-shaped cleft palate, with airway obstruction and feeding difficulty. **Stickler syndrome** is the commonest syndrome underlying Robin sequence; it is a collagen disorder with high myopia, *retinal detachment* (so eye screening is essential), hearing loss and joint problems. The **22q11.2 deletion (velocardiofacial or DiGeorge) syndrome** brings cleft palate or velopharyngeal insufficiency with heart defects, hypocalcaemia, immune deficiency and learning difficulty, and its carotids may lie *medially* behind the pharynx, a danger at pharyngeal surgery. Treacher Collins syndrome, hemifacial microsomia, the CHARGE association and Apert syndrome complete the list.

2 CHAPTER TWO Classification

Describing a cleft.

2.1 Lip and palate

A cleft lip is unilateral or bilateral. It is *complete* when it runs through the nasal sill, *incomplete* when the sill is intact, sometimes with a *Simonart band* of tissue bridging the gap, and *microform* when there is only a notch and a muscle gap under intact skin.

A cleft palate is *unilateral* when one palatal shelf remains attached to the vomer and *bilateral* when the vomer lies free in the midline. Veau's classification grades the extent: group I involves the soft palate only, group II the soft and hard palate up to the incisive foramen, group III is a complete unilateral cleft through lip, alveolus and palate, and group IV a complete bilateral cleft.

2.2 The submucous cleft

In a submucous cleft the mucosa is intact but the muscles are separated beneath it. The classic signs form *Calnan's triad*: a *bifid uvula*, a *zona pellucida* (a translucent bluish strip in the midline of the soft palate), and a *notch in the posterior edge of the hard palate* that can be felt with a finger. An *occult* submucous cleft has none of these

signs and is seen only on nasendoscopy, as a missing muscle bulge on the nasal surface of the velum.

CHECK THE PALATE BEFORE ADENOIDECTOMY

A bifid uvula, a palpable notch or hypernasal speech means a possible submucous cleft or velopharyngeal weakness. A full adenoidectomy can unmask *velopharyngeal insufficiency*, because the adenoid was helping the palate seal. Avoid adenoidectomy, or do a limited (superior) one, and involve the cleft team.

3 CHAPTER THREE Early care

Feeding, airway, ears.

3.1 The newborn

Many clefts are now diagnosed by antenatal ultrasound, which allows the family to be counselled before birth. After birth, feeding is the first problem: a baby with a cleft palate cannot create suction, so breastfeeding is usually impossible except with a cleft lip alone. Squeezy bottles or special teats such as the Haberman or pigeon, with feeding upright, are used, and weight gain is watched closely.

The airway matters most in Robin sequence. Prone or side positioning and a nasopharyngeal airway manage most babies; severe obstruction needs a tongue–lip adhesion, *mandibular distraction osteogenesis* or a tracheostomy, guided by sleep studies. Every baby is examined for other anomalies, especially cardiac and syndromic, reviewed by a geneticist, and given a *hearing screen*. Presurgical orthopaedics, by lip taping or *nasoalveolar moulding*, narrows a wide cleft and shapes the nose before lip repair.

3.2 The ears

Because the tensor and levator insert abnormally, the Eustachian tube cannot open, and otitis media with effusion affects *almost every child (over 90%)* with a cleft palate. Hearing is tested from birth and throughout childhood, and persistent hearing loss is treated with ventilation tubes or hearing aids, the tubes often being

inserted at the time of lip or palate repair. The ears usually improve after palate repair and with growth, but the long-term risks of chronic otitis media, retraction pockets and cholesteatoma remain.

OPERATIVE GUIDES IN THEATRE 1 operation

Mandibular distraction.

Mandibular Distraction
Mandibular distraction

4 CHAPTER FOUR Surgery

To a timetable.

4.1 The timetable

A TYPICAL CLEFT CARE TIMELINE

AGE	STEP
Antenatal and birth	Diagnosis, counselling, feeding plan, airway, hearing screen, genetics
Weeks 1 to 12	Presurgical orthopaedics (lip taping, nasoalveolar moulding) if needed
About 3 months	Cleft lip repair, with or without primary rhinoplasty (traditional "rule of 10s": 10 weeks, 10 lb, haemoglobin 10 g/dL)
6 to 12 (up to 18) months	Cleft palate repair, with ear tubes as needed
3 to 5 years	Speech assessment; velopharyngeal surgery if needed; fistula repair
8 to 11 years (mixed dentition, before the canine erupts)	Alveolar bone graft (cancellous bone from the iliac crest)
Teens	Orthodontics; orthognathic surgery (maxillary advancement) after growth

AGE	STEP
16 to 18 years	Definitive septorhinoplasty; lip revision

4.2 Cleft lip repair

Lip repair has four aims: to join the *orbicularis oris* into a working sphincter, to align the white roll and vermilion exactly, to create a symmetrical philtrum, nostril sill and alar base, and to correct the nose.

The unilateral cleft lip nose has a characteristic deformity. The cleft-side alar cartilage is displaced down, back and out with a flat, low dome, the alar base is wide, the columella and caudal septum deviate to the *non-cleft* side, and the maxilla on the cleft side is hypoplastic.

In the **Millard rotation–advancement repair**, the *non-cleft (medial) side* is *rotated* down to level Cupid's bow, and the *cleft (lateral) side* is *advanced* into the gap beneath the nostril. The muscle is released from its abnormal insertions and repaired across the cleft, and the nasal floor is closed. The Fisher anatomical subunit repair and triangular-flap repairs are the alternatives.

The **bilateral cleft lip** is harder, because of a protruding premaxilla, a prolabium without muscle and a short columella; it is repaired in one or two stages, building the muscle across behind the prolabium. The complications of lip repair are dehiscence, a vermilion notch (the "whistle deformity"), a mismatched white roll, scar hypertrophy and nasal asymmetry.

4.3 Cleft palate repair

The palate is repaired at about 9 to 12 months, and certainly before 18 months. The timing balances *better speech with early repair* against *better maxillary growth with later repair*. The aims are to separate the mouth from the nose and, above all, to *rebuild the levator sling* by intravelar veloplasty and lengthen the palate for normal speech, because in a cleft the levator runs front to back and inserts into the back of the hard palate instead of meeting its partner in the midline.

The **two-flap palatoplasty (Bardach)** raises mucoperiosteal flaps on the greater palatine pedicles, closes the nasal and oral layers and adds an intravelar veloplasty;

it is used for complete clefts. The **von Langenbeck repair** uses bipedicled flaps with lateral relaxing incisions. The **Furlow double-opposing Z-plasty** places opposite Z-plasties on the oral and nasal layers, which *lengthens the palate and realigns the levator*, and is used for soft palate clefts, submucous clefts and velopharyngeal insufficiency. The older **Veau–Wardill–Kilner V–Y pushback** leaves more raw bone and restricts growth.

After repair, bleeding and airway obstruction, especially in Robin sequence, are the early risks. An *oronasal fistula* develops in up to about 10%, usually at the junction of hard and soft palate, *velopharyngeal insufficiency* persists in up to 20 to 25%, and maxillary hypoplasia appears with growth. A submucous cleft is repaired only if it causes velopharyngeal insufficiency or feeding or ear problems, by Furlow palatoplasty or intravelar veloplasty.

OPERATIVE GUIDES IN THEATRE 2 operations

Cleft palate and velopharyngeal dysfunction, and surgery of the soft palate.

Cleft Palate and Velopharyngeal Dysfunction
Cleft palate and VPI

Surgery of the Soft Palate
Soft palate

5 CHAPTER FIVE Velopharyngeal dysfunction

Speech and the palate.

5.1 Definitions

Three kinds of velopharyngeal dysfunction are distinguished. **Insufficiency** is a structural deficit: the palate is too short or scarred, there is a cleft or submucous cleft, or tissue has been lost after adenoidectomy or tumour surgery. **Incompetence** is a neuromuscular deficit, as in palsies or the 22q11 deletion. **Mislearning** is an articulation pattern that needs speech therapy only.

5.2 Features

The child has *hypernasal* resonance on vowels and voiced oral consonants, while the nasal sounds /m/, /n/ and /ng/ remain normal. *Nasal air emission* and turbulence accompany pressure consonants such as p, b, t, d and s; consonants are weak; *compensatory articulations* such as glottal stops develop; and food and liquid may regurgitate through the nose.

5.3 Assessment

A speech and language therapist makes a perceptual assessment, and the velopharynx is imaged during speech by *nasendoscopy*, which shows the size and site of the gap and the closure pattern, and by *multiview videofluoroscopy*. The closure pattern may be coronal (the commonest), sagittal, circular, or circular with a Passavant ridge, and it determines the operation. Before any pharyngeal surgery the child is checked for the 22q11 deletion with *medialised carotids*, ideally by MRA, and for obstructive sleep apnoea.

5.4 Treatment

Surgery corrects structure and therapy corrects articulation, so speech therapy alone is used for mislearning and small gaps. For a short palate or malpositioned muscle with a small gap, the palate is re-repaired with a Furlow double-opposing Z-plasty or a radical intravelar veloplasty. Small central gaps can be closed by *augmenting the posterior pharyngeal wall* with fat, implants or fillers, at the risk of migration and extrusion.

For a **large central gap with good lateral wall movement**, a *superiorly based pharyngeal flap* is raised from the posterior wall and attached to the velum, leaving two lateral ports. For a **circular or bow-tie gap with poor lateral wall movement**, a *sphincter pharyngoplasty* (Orticochea, Jackson or Hynes) transposes the posterior tonsillar pillars with palatopharyngeus and sutures them to the posterior wall, narrowing the port while leaving a central opening. A speech bulb or palatal lift helps poor surgical candidates.

The main risks of pharyngeal surgery are *obstructive sleep apnoea* and hyponasality, so oxygen saturation is monitored after the operation.

OPERATIVE GUIDES IN THEATRE 1 operation

Advanced palatal surgery: expansion sphincter and transpalatal advancement pharyngoplasty.

Advanced Palatal Surgery: Expansion Sphincter Pharyngoplasty & Transpalatal Advancement Pharyngoplasty
Pharyngoplasty

6 CHAPTER SIX Later care

Bone, jaws, nose.

6.1 Through growth

The alveolar cleft is grafted in the mixed dentition, at about 8 to 11 years and before the permanent canine erupts, with cancellous bone from the iliac crest. The graft closes the alveolar cleft and any oronasal fistula there and supports the teeth and the alar base.

About a quarter of children develop maxillary hypoplasia with a class III bite and need a *Le Fort I advancement* or distraction once growth stops. The cleft lip nose, corrected in part by a primary rhinoplasty at lip repair, is finally treated by a definitive *septorhinoplasty* with cartilage grafts after facial growth. Dental care and orthodontics continue throughout, and psychological support for the child and family matters most at school age and in adolescence.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Primary versus secondary palate;** the incisive foramen; the embryology of each.

- **The nerve supply of the palatal muscles** (tensor by V3).
- **Cleft lip versus isolated cleft palate:** epidemiology and syndromes (Van der Woude, Robin, Stickler, 22q11).
- **The submucous cleft triad,** and why it matters before adenoidectomy.
- **The timetable of care,** and the rule of 10s.
- **Millard repair:** which side rotates.
- **Furlow palatoplasty and intravelar veloplasty.**
- **Otitis media with effusion in cleft palate:** why, and how it is managed.
- **Velopharyngeal insufficiency:** signs, nasendoscopy, pharyngeal flap versus sphincter pharyngoplasty, OSA risk.

7.2 Common pitfalls

COMMON PITFALLS

- Adenoidectomy in a child with an unrecognised submucous cleft.
- Missing Stickler syndrome (retinal detachment) in Robin sequence.
- Pharyngeal surgery in 22q11 deletion without imaging the carotids.
- Neglecting hearing surveillance after the palate is repaired.

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IMAGING

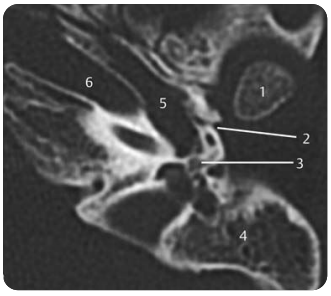
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · CONGENITAL

Malformations, Treacher Collins

- In deformities of the temporal bone, pay particular attention to the:
- External auditory canal, bony or fibrous atresia, appearance and position of the mandibular condyle and temporomandibular joint.
- Appearance of the middle ear cavity and mastoid pneumatization.
- Signs of ankylosis or disconnections of the ossicular chain.
- Presence of inner ear deformities, the round and oval windows, and the vestibular aqueduct.
- Aberrations in the course and anterior displacement of the facial nerve, which may complicate surgery.



CT, axial. For orientation, this is a slice through the basal cochlear turn and the mandibular condyle (1). Behind the condyle, the bony part of the external…